

# Byung-Kwan Lee

Ph.D., School of Electrical Engineering at KAIST

Research Scientist at  NVIDIA

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## Research Interest

Building efficient, high-performing Vision-Language Models (VLMs), with focus on:

- Distillation / RL
- Pruning
- Compression
- Data / Training / Inference Efficiency

## Work Experience

### NVIDIA • Research Scientist

Oct. 2025 - Current

**LEAD** Masking Teacher and Reinforcing Student [COMPLETED] Mask-progressive RL distillation that gradually unmask teacher weights and uses offline RL with accuracy & distillation rewards.

CVPR 2026 Accept U.S. Non-Provisional Patent Filed ArXiv Release Internal Release

**LEAD** GenRecal [COMPLETED] Cross-architecture VLM distillation via a Recalibrator that aligns heterogeneous token representations regardless of vocabulary, token splits, or index ordering.

Under Review U.S. Patent Upgraded to Non-Provisional Internal Tech Transfer

**LEAD** Unified RL & Imitation Learning for VLMs [COMPLETED] Combines RL with adversarial imitation, using an LLM-based discriminator and multi-teacher guidance to build lightweight yet powerful VLMs.

NeurIPS 2025 Accept U.S. Patent Upgraded to Non-Provisional ArXiv Release Internal Release

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### NVIDIA • Research Intern

Oct. 2024 - Oct. 2025

**LEAD** VLsI: Verbalized Layers-to-Interactions [COMPLETED] Layer-wise distillation using intermediate verbalizers, enabling small VLMs (2B/7B) to align with large VLMs' reasoning progression and outperform GPT-4V.

CVPR 2025 Accept U.S. Non-Provisional Patent Filed ArXiv Release Internal Release Internal Tech Transfer

**LEAD** GenRecal [INITIATED] Initial design of the Recalibrator framework for cross-tokenizer VLM distillation; first proof-of-concept and internal demo.

U.S. Provisional Patent Filed ArXiv Release Internal Release

**LEAD** Unified RL & Imitation Learning for VLMs [INITIATED] First formulation of the RL + adversarial imitation training pipeline; initial experiments and team setup.

U.S. Provisional Patent Filed

## Education

### KAIST

Mar. 2020 - Aug. 2025

Ph.D., School of Electrical Engineering

◦ GPA: 3.77/4.3

◦ Dissertation: Building High-performing, Efficient-size Vision Language Models: Merge, Modify, and Distill [Link]

### KAIST

Mar. 2018 - Feb. 2020

M.S., The Cho Chun Sik Graduate School of Green Transportation

◦ GPA: 3.72/4.3

◦ Thesis: Training Encoder-Attention through Fully-Connected CRFs for Efficient End-to-End Lane Detection Model [Link]

### Hanyang University

Mar. 2014 - Feb. 2018

B.S., Mathematics and Electronic Engineering

◦ GPA: 3.86/4.5

## Publications

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|                  |                                         |
|------------------|-----------------------------------------|
| Overall          | 16 Accepts · 3 Pending · 2 Tech Reports |
| Computer Vision  | 5 CVPR · 2 ICCV · 1 ECCV · 1 ICIP       |
| Machine Learning | 3 NeurIPS · 1 ICLR                      |
| NLP              | 1 ACL · 1 EMNLP                         |
| Journal          | 1 Pattern Recognition                   |

1. **“Hide to See: Reasoning-prefix Masking for Visual-anchored Thinking in VLM Distillation”**  
Seonghoon Yu, Dongjun Nam, Byung-Kwan Lee, Jeany Son  
Under Review [[Paper](#)][[Code](#)]
2. **“Why and When Visual Token Pruning Fails? A Study on Relevant Visual Information Shift in MLLMs Decoding”**  
Jiwan Kim, Kibum Kim, Wonjoong Kim, Byung-Kwan Lee, Chanyoung Park  
Under Review [[Paper](#)][[Project](#)]
3. **“GenRecal: Generation after Recalibration from Large to Small Vision Language Models”**  
Byung-Kwan Lee, Ryo Hachiuma, Yong Man Ro, Yu-Chiang Frank Wang, and Yueh-Hua Wu  
Under Review, [[Paper](#)][[Project](#)]  
U.S. Patent Application Filed (Non-Provisional), NVIDIA Research, 2025
4. **“Masking Teacher and Reinforcing Student for Distilling Vision-Language Models”**  
Byung-Kwan Lee, Yu-Chiang Frank Wang, Ryo Hachiuma  
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026 [[Paper](#)]  
U.S. Patent Application Filed (Non-Provisional), NVIDIA Research, 2026
5. **“Recursive Think-Answer Process for LLMs and VLMs”**  
Byung-Kwan Lee\*, Youngchae Chee\*, Yong Man Ro  
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings, 2026 [[Paper](#)][[Project](#)]
6. **“RefineBench: Evaluating Refinement Capability in Language Models”**  
Young-Jun Lee\*, Seungone Kim\*, Byung-Kwan Lee, Minkyong Moon, Yechan Hwang, Jong Myoung Kim, Graham Neubig, Sean Welleck, Ho-Jin Choi  
International Conference on Learning Representations (ICLR), 2026 [[Paper](#)][[Project](#)]  
Best Runner-Up Award, Workshop for Multi-Turn Interactions in Large Language Models, NeurIPS 2025 (Oral, Top 1%) [[Link](#)]
7. **“Unified Reinforcement and Imitation Learning for Vision-Language Models”**  
Byung-Kwan Lee, Ryo Hachiuma, Yong Man Ro, Yu-Chiang Frank Wang, Yueh-Hua Wu  
Neural Information Processing Systems (NeurIPS), 2025 [[Paper](#)][[Project](#)]  
U.S. Patent Application Filed (Non-Provisional), NVIDIA Research, 2025
8. **“MultiVerse: A Multi-Turn Conversation Benchmark for Evaluating Large Vision and Language Models”**  
Young-Jun Lee, Byung-Kwan Lee, Jianshu Zhang, Yechan Hwang, Byungsoo Ko, Han-Gyu Kim, Dongyu Yao, Xuankun Rong, Eojin Joo, Seung-Ho Han, Bowon Ko, Ho-Jin Choi  
IEEE/CVF International Conference on Computer Vision (ICCV), 2025 [[Paper](#)][[Project](#)]  
Workshop for Knowledge-Intensive Multimodal Reasoning, ICCV 2025 [[Link](#)]
9. **“VLsI: Verbalized Layers-to-Interactions from Large to Small Vision Language Models”**  
Byung-Kwan Lee, Ryo Hachiuma, Yu-Chiang Frank Wang, Yong Man Ro, Yueh-Hua Wu  
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025 [[Paper](#)][[Project](#)]  
U.S. Patent Application Filed (Non-Provisional), NVIDIA Research, 2025

10. **“Phantom of Latent for Large Language and Vision Models”**  
Byung-Kwan Lee, Sangyun Chung, Chae Won Kim, Beomchan Park, Yong Man Ro  
Technical Report [[Paper](#)][[Code](#)][[Huggingface Model](#)]
11. **“TroL: Traversal of Layers for Large Language and Vision Models”**  
Byung-Kwan Lee, Sangyun Chung, Chae Won Kim, Beomchan Park, Yong Man Ro  
Empirical Methods in Natural Language Processing (EMNLP), 2024 [[Paper](#)][[Code](#)][[Huggingface Model](#)]
12. **“Meteor: Mamba-based Traversal of Rationale for Large Language and Vision Models”**  
Byung-Kwan Lee, Chae Won Kim, Beomchan Park, Yong Man Ro  
Neural Information Processing Systems (NeurIPS), 2024 [[Paper](#)][[Code](#)][[Huggingface Model](#)]  
31st Samsung HumanTech Paper Awards in section of Computer Science & Engineering
13. **“MoAI: Mixture of All Intelligence for Large Language and Vision Models”**  
Byung-Kwan Lee, Beomchan Park, Chae Won Kim, Yong Man Ro  
European Conference on Computer Vision (ECCV), 2024, [[Paper](#)][[Code](#)][[Huggingface Model](#)]
14. **“CoLLaVO: Crayon Large Language and Vision mOdel”**  
Byung-Kwan Lee, Beomchan Park, Chae Won Kim, Yong Man Ro  
Findings of the Association for Computational Linguistics (ACL), 2024, [[Paper](#)][[Code](#)][[Huggingface Model](#)]  
2024 KCC XAI Workshop, Best Paper Awards
15. **“Causal Unsupervised Semantic Segmentation”**  
Junho Kim\*, Byung-Kwan Lee\*, and Yong Man Ro  
Journal of Pattern Recognition [[Paper](#)][[Code](#)]
16. **“Mitigating Adversarial Vulnerability through Causal Parameter Estimation by Adversarial Double Machine Learning”**  
Byung-Kwan Lee\*, Junho Kim\*, and Yong Man Ro  
IEEE/CVF International Conference on Computer Vision (ICCV), 2023 [[Paper](#)][[Code](#)]
17. **“Mitigating Dataset Bias in Image Captioning through CLIP Confounder-free Captioning Network”**  
YeonJu Kim, Junho Kim, Byung-Kwan Lee, Sebin Shin, and Yong Man Ro  
IEEE International Conference on Image Processing (ICIP), 2023 [[Paper](#)][[Code](#)]
18. **“Demystifying Causal Features on Adversarial Examples and Causal Inoculation for Robust Network by Adversarial Instrumental Variable Regression”**  
Junho Kim\*, Byung-Kwan Lee\*, and Yong Man Ro  
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023 [[Paper](#)][[Code](#)]
19. **“Masking Adversarial Damage: Finding Adversarial Saliency for Robust and Sparse Network”**  
Byung-Kwan Lee\*, Junho Kim\*, and Yong Man Ro  
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022 [[Paper](#)][[Code](#)]
20. **“Distilling Robust and Non-Robust Features in Adversarial Examples by Information Bottleneck”**  
Junho Kim\*, Byung-Kwan Lee\*, and Yong Man Ro  
Neural Information Processing Systems (NeurIPS), 2021 [[Paper](#)][[Code](#)]
21. **“Towards Adversarial Robustness of Bayesian Neural Network through Hierarchical Variational Inference”**  
Byung-Kwan Lee\*, Youngjun Yu, and Yong Man Ro  
Technical Report [[Paper](#)][[Code](#)]

## Reviewer Experience

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)

*Journal*

IEEE Transactions on Neural Networks and Learning Systems (TNNLS)

*Journal*

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|------------------------------------------------------------------------|------------|
| IEEE Transactions on Circuits and Systems for Video Technology (TCSVT) | Journal    |
| IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)  | Conference |
| International Conference on Computer Vision (ICCV)                     | Conference |
| European Conference on Computer Vision (ECCV)                          | Conference |
| International Conference on Learning Representations (ICLR)            | Conference |
| Neural Information Processing Systems (NeurIPS)                        | Conference |
| International Conference on Machine Learning (ICML)                    | Conference |
| AAAI Conference on Artificial Intelligence (AAAI)                      | Conference |
| Annual Meeting of the Association for Computational Linguistics (ACL)  | Conference |
| Conference on Empirical Methods in Natural Language Processing (EMNLP) | Conference |

## Invited Talks and Awards

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|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Silver Reviewer Award, International Conference on Machine Learning (ICML)                                                              | 2026                  |
| Best Runner-Up Award (Oral, Top 1%), Multi-Turn Interactions in LLMs Workshop @ NeurIPS                                                 | 2025                  |
| Invited Talk in Kongju University                                                                                                       | 2025                  |
| Invited Talk in Hanyang University ERICA                                                                                                | 2025                  |
| Invited Talk in Kookmin University                                                                                                      | 2025                  |
| Invited Talk in Dongseo University                                                                                                      | 2025                  |
| 31st Samsung HumanTech Paper Awards in section of Computer Science & Engineering                                                        | 2025                  |
| KCC XAI Workshop, Best Paper Awards                                                                                                     | 2024                  |
| Invited Talk in Korea Institute of Science and Technology Information (KISTI)                                                           | 2024                  |
| Invited Talk in NAVER HyperCLOVA X                                                                                                      | 2024                  |
| Invited Talk in Kakao Brain                                                                                                             | 2024                  |
| 1st Award for Research Presentation in Center for Applied Research in Artificial Intelligence (CARAI)                                   | 2023                  |
| GIRE Research Mentor, Seoul International School – “MUSE”, Journal of Student Research [ <a href="#">Publication</a> ]                  | 2023                  |
| KAIST SW IT Academy, AI Project Director of 1st Award Topic                                                                             | 2023                  |
| Invitation to Presenter in Korean Conference on Computer Vision (KCCV)                                                                  | 2023                  |
| KAIST SW IT Academy, Lecturer of Data Structure & Algorithm, Deep Learning, Computer Vision                                             | 2023                  |
| Invitation to Presenter in Korean Conference on Computer Vision (KCCV)                                                                  | 2022                  |
| 2nd Award for Research Presentation in Center for Applied Research in Artificial Intelligence (CARAI)                                   | 2022                  |
| KAIST Fellowship                                                                                                                        | Mar. 2020 - Feb. 2024 |
| National Government Fellowship                                                                                                          | Mar. 2018 - Feb. 2020 |
| 1st President's Award for International Autonomous Vehicle Competition, Team Leader [ <a href="#">YouTube</a> ][ <a href="#">Talk</a> ] | 2018                  |